Week 7 - Dry and Moist Air: Worked Answers

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS**: selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY**: expected independent practice after the session.
- **EXTRA CHALLENGE**: optional problems for students who want additional depth or a harder application.

The categories are repeated here so the worked answers map directly onto the practice sheet. Use the solutions to check method, units and physical interpretation rather than only the final number. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

Useful constants: $R_v = 461.5\ \mathrm{J\,kg^{-1}\,K^{-1}}$, $\epsilon = R_d/R_v \approx 0.622$, $L_v \approx 2.5\times10^6\ \mathrm{J\,kg^{-1}}$. A useful estimate for lifting-condensation-level height is $z_{LCL} \approx 125(T - T_d)$ metres when T and T_d are in °C. Approximate saturation vapour pressure over liquid water:

T (°C)	-10	0	5	10	15	20	25
e_s (hPa)	2.87	6.11	8.72	12.27	17.04	23.37	31.67

YOU SHOULD TRY Question 1: Phase changes and atmospheric energy

For evaporation, condensation, melting, freezing, sublimation and deposition:

1. state the direction of the phase change;
2. state whether latent heat is absorbed from or released to the surroundings;
3. identify which two of these processes are especially important when cloud water or ice forms from water vapour.

Worked answer

Evaporation is liquid → vapour and absorbs latent heat. Condensation is vapour → liquid and releases latent heat. Melting is solid → liquid and absorbs latent heat. Freezing is liquid → solid and releases latent heat. Sublimation is solid → vapour and absorbs latent heat. Deposition is vapour → solid and releases latent heat.

Condensation and deposition are especially important during cloud formation because they convert atmospheric water vapour into liquid droplets or ice and release latent heat into the air.

IN CLASS Question 2: Relative humidity and dew point

At 25°C, an air parcel has water-vapour partial pressure $e = 17.0$ hPa. Relative humidity is defined by

$$RH = 100\frac{e}{e_s(T)},$$

where $e_s(T)$ is the saturation vapour pressure at the air temperature.

1. Calculate the relative humidity.

2. Use the table to estimate the dew-point temperature.
3. Explain physically what would happen if the parcel were cooled below its dew point while its water-vapour content initially remained unchanged.

Worked answer

At 25°C, $e_s = 31.67$ hPa. Therefore

$$RH = 100\frac{e}{e_s} = 100\frac{17.0}{31.67} = 53.7\%.$$

The dew point is the temperature at which $e_s(T_d) = e$. The table gives $e_s \approx 17.04$ hPa at 15°C, so

$$T_d \approx 15°\text{C}.$$

If the air is cooled below this temperature, the saturation vapour pressure becomes smaller than the existing vapour pressure. The air becomes supersaturated and condensation onto available nuclei occurs, reducing the vapour content toward saturation.

EXTRA CHALLENGE Question 3: Water-vapour density

For the air in Question 2, calculate the density of water vapour using

$$e = \rho_v R_v T.$$

Worked answer

Convert $e = 17.0$ hPa to 1700 Pa and $T = 25°\text{C}$ to 298.15 K:

$$\rho_v = \frac{e}{R_v T} = \frac{1700}{461.5 \times 298.15} = 1.24 \times 10^{-2}\ \text{kg m}^{-3}.$$

Thus the water-vapour density is about 0.0124 kg m^{-3}, or 12.4 g m^{-3}.

IN CLASS Question 4: Mixing ratio and specific humidity

The **water-vapour mixing ratio** is the mass of water vapour per unit mass of dry air,

$$r_v = \frac{m_w}{m_d},$$

whereas the **specific humidity** is the mass of water vapour per unit mass of moist air,

$$q = \frac{m_w}{m_w + m_d}.$$

At total pressure $p = 1000$ hPa and vapour pressure $e = 17.0$ hPa:

1. Calculate the mixing ratio using
$$r_v = \epsilon\frac{e}{p - e}.$$
2. Starting from the definitions above, show that
$$q = \frac{r_v}{1 + r_v},$$
and calculate q.
3. Give both r_v and q in g kg^{-1} and explain why they are numerically similar for ordinary atmospheric air.

Worked answer

The water-vapour mixing ratio is the mass of water vapour per unit mass of dry air, so

$$r_v = \frac{m_w}{m_d}.$$

Using the pressure form,

$$r_v = 0.622\frac{17.0}{1000 - 17.0} = 0.01075 \text{ kg kg}^{-1} = 10.75 \text{ g kg}^{-1}.$$

The specific humidity is the mass of water vapour per unit mass of moist air,

$$q = \frac{m_w}{m_w + m_d}.$$

Since $r_v = m_w/m_d$, we have $m_w = r_v m_d$. Therefore

$$q = \frac{r_v m_d}{r_v m_d + m_d} = \frac{r_v}{1 + r_v}.$$

Hence

$$q = \frac{0.01075}{1.01075} = 0.01064 \text{ kg kg}^{-1} = 10.64 \text{ g kg}^{-1}.$$

They are similar because atmospheric water vapour is usually only a small fraction of the total mass, so $r_v \ll 1$ and therefore $1 + r_v \approx 1$.

YOU SHOULD TRY Question 5: Why moist air is less dense than dry air

A mixture contains 98% dry air and 2% water vapour by mole fraction. Take molar masses $M_d = 28.97 \text{ g mol}^{-1}$ and $M_v = 18.02 \text{ g mol}^{-1}$.

1. Calculate the mean molar mass of the mixture.
2. At the same pressure and temperature, would this mixture be denser or less dense than dry air? Explain.

Worked answer

The mole-fraction-weighted molar mass is

$$M = 0.98(28.97) + 0.02(18.02) = 28.75 \text{ g mol}^{-1}.$$

This is smaller than the molar mass of dry air. In density form the ideal-gas law is

$$p = \rho RT,$$

where the specific gas constant is related to molar mass by

$$R = \frac{R^*}{M}.$$

Thus, at fixed pressure and temperature,

$$\rho = \frac{p}{RT} = \frac{pM}{R^*T},$$

so density is proportional to molar mass. Because adding water vapour lowers the mean molar mass, moist air is slightly less dense than dry air at the same p and T.

IN CLASS Question 6: Estimating cloud-base height

For an unsaturated parcel lifted approximately adiabatically, take the parcel temperature to decrease at the dry-adiabatic lapse rate,

$$T(z) = T_0 - \Gamma_d z, \qquad \Gamma_d \approx 9.8 \text{ K km}^{-1}.$$

The parcel dew-point temperature also decreases as the parcel expands and its vapour pressure falls; a useful approximation is

$$T_d(z) = T_{d0} - \Gamma_{dew} z, \qquad \Gamma_{dew} \approx 1.8 \text{ K km}^{-1}.$$

The lifting condensation level (LCL) is reached when $T = T_d$.

For surface temperature $T_0 = 22°\text{C}$ and dew point $T_{d0} = 14°\text{C}$:

1. Show that
$$z_{LCL} \approx \frac{T_0 - T_{d0}}{\Gamma_d - \Gamma_{dew}},$$
and hence recover the familiar approximation $z_{LCL} \approx 125(T_0 - T_{d0})$ m when temperatures are in °C.
2. Estimate the LCL height.
3. Explain physically why a larger surface $T_0 - T_{d0}$ difference usually implies a higher cloud base for a lifted parcel.

Worked answer

At the LCL the parcel temperature and dew point are equal:

$$T_0 - \Gamma_d z_{LCL} = T_{d0} - \Gamma_{dew} z_{LCL}.$$

Therefore

$$z_{LCL} = \frac{T_0 - T_{d0}}{\Gamma_d - \Gamma_{dew}}.$$

Using $\Gamma_d - \Gamma_{dew} = 9.8 - 1.8 = 8.0 \text{ K km}^{-1}$ gives

$$z_{LCL} = \frac{T_0 - T_{d0}}{8.0} \text{ km} = 0.125(T_0 - T_{d0}) \text{ km},$$

or

$$\boxed{z_{LCL} \approx 125(T_0 - T_{d0}) \text{ m}}.$$

For $T_0 - T_{d0} = 8°\text{C}$,

$$z_{LCL} \approx 125 \times 8 = 1000 \text{ m}.$$

The dew point falls with height because expansion lowers the parcel pressure and hence the water-vapour partial pressure. However, an unsaturated parcel temperature falls much faster, at approximately the dry-adiabatic lapse rate. A larger initial $T_0 - T_{d0}$ therefore requires a greater ascent before the two temperatures meet and condensation begins.

EXTRA CHALLENGE Question 7: A psychrometer

A ventilated psychrometer has a dry-bulb thermometer and a wet-bulb thermometer.

1. Why is the wet bulb usually colder than the dry bulb when the air is unsaturated?
2. What happens to the wet-bulb depression as relative humidity approaches 100%?
3. Why is ventilation important?

Worked answer

Water evaporating from the wet wick requires latent heat. That energy is taken from the thermometer/water/air immediately around the wet bulb, cooling it below the dry-bulb temperature. As relative humidity approaches 100%, evaporation becomes progressively weaker, so the wet-bulb depression tends toward zero. Ventilation continually replaces air near the wick and makes the evaporation rate representative and reproducible rather than allowing a locally saturated layer to form around the sensor.

YOU SHOULD TRY Question 8: Latent heating by condensation

A rising cloud parcel condenses 2.0 kg of water. Estimate the latent heat released. If that energy were used only to warm 1000 kg of air with $c_p = 1004\ \mathrm{J\,kg^{-1}\,K^{-1}}$, what temperature increase would it correspond to?

Worked answer

The released latent heat is

$$Q = mL_v = 2.0 \times 2.5 \times 10^6 = 5.0 \times 10^6\ \mathrm{J}.$$

If all this energy heated 1000 kg of air,

$$\Delta T = \frac{Q}{mc_p} = \frac{5.0 \times 10^6}{1000 \times 1004} = 4.98\ \mathrm{K}.$$

The real atmosphere does not channel all latent heat into a single isolated air mass in this way, but the calculation illustrates the large energetic effect of condensation.